

15.1: The Probability of Events from 2 Experiments:

If 2 events are independent:

$\Rightarrow P(A \cap B) = P(A)P(B)$

15.2: Conditional Probabilities:

If 2 events are independent $\Rightarrow P(A|B)$ (probability of A, given B has occured) $= P(A)$. Each event is unaffected.
Hence:

$P(A \cap B) \equiv P(A)P(B|A)$

$\Rightarrow P(B|A) = \frac{P(A \cap B)}{P(A)}$

Hence:

$P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$

$\Rightarrow P(A|B) = \frac{P(A)P(B|A)}{P(B)}$

Selections: A Recap:

For fully random events, of N total outcomes, n picked, K total desired, k desired wanted:

$$P(E) = \frac{{}^{k}C_{k} \times {}^{N-k}C_{n-k}}{{}^{N}C_{n}}$$